Recent Advances in the Foundations and Applications of Unbiased Learning to Rank

Shashank Gupta
University of Amsterdam
Amsterdam, The Netherlands
s.gupta2@uva.nl

Philipp Hager
University of Amsterdam
Amsterdam, The Netherlands
p.k.hager@uva.nl

Jin Huang
University of Amsterdam
Amsterdam, The Netherlands
j.huang2@uva.nl

Ali Vardasbi
University of Amsterdam
Amsterdam, The Netherlands
a.vardasbi@uva.nl

Harrie Oosterhuis
Radboud University
Nijmegen, The Netherlands
harrie.oosterhuis@ru.nl

COVER SHEET

1 TITLE

Recent Advances in the Foundations and Applications of Unbiased Learning to Rank

2 LENGTH

The proposed length of the tutorial is a half-day block of 3 hours, in addition to breaks.

3 TUTORIAL FORMAT

The tutorial will be delivered on-site. All presenters commit to presenting the tutorial in-person.

4 INTENDED AUDIENCE

This tutorial is intended for academics and practitioners interested in unbiased learning to rank. It will be valuable to those without prior knowledge of the topic and those with intermediate knowledge who wish to learn about the latest advancements in the field. Therefore, previous knowledge of unbiased learning to rank is not required, but a prior understanding of basic learning to rank concepts is expected.

5 DETAILED CONTACT INFORMATION

- **Shashank Gupta** (main contact person) Informatics Institute, University of Amsterdam, Science Park 904, 1098 XH Amsterdam, The Netherlands. Email: s.gupta2@uva.nl. Contact no: +31-643225590.
- **Philipp Hager** Informatics Institute, University of Amsterdam, Science Park 904, 1098 XH Amsterdam, The Netherlands. Email: p.k.hager@uva.nl.
- **Jin Huang** Informatics Institute, University of Amsterdam, Science Park 904, 1098 XH Amsterdam, The Netherlands. Email: j.huang2@uva.nl.
- **Ali Vardasbi** Informatics Institute, University of Amsterdam, Science Park 904, 1098 XH Amsterdam, The Netherlands. Email: a.vardasbi@uva.nl.
- **Harrie Oosterhuis** Faculty of Science, University of Nijmegen, Postbus 9010, 6500 GL Nijmegen, The Netherlands. Email: harrie.oosterhuis@ru.nl.

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6 BRIEF BIOGRAPHIES

6.1 Shashank Gupta

Shashank Gupta is a second-year PhD student at the University of Amsterdam (UvA) working under the supervision of Prof. Dr. Maarten de Rijke and Dr. Harrie Oosterhuis. His main research interests are counterfactual evaluation and learning from user interactions. Before joining UvA, he worked as a data scientist with the search team at Flipkart, India's largest e-commerce company. At Flipkart, he worked on various applied IR problems, such as personalized search, spelling correction, query classification, and session length prediction [17]. He completed his MS by Research at IIIT-Hyderabad, India.

Shashank has published on the topics of medical social media mining [15, 16, 19], news recommendation [35, 37–39], fake news detection [20], hate speech detection [4, 21], and bias in recommender systems [18]. He got the best poster award at the WWW'17 for his work on hate speech detection [4]. Shashank has teaching experience at the UvA, where contributes to the Advanced Information Retrieval course. Previously, he has also contributed to the Information Retrieval and Machine Learning courses at IIIT-Hyderabad, India. Recently, he has presented an invited talk on counterfactual evaluation and learning at Huawei, Amsterdam.

6.2 Philipp Hager

Philipp Hager is a second-year PhD candidate at the University of Amsterdam (UvA) supervised by Prof. Dr. Maarten de Rijke and Dr. Onno Zoeter at Booking.com. His research interest includes offline learning and evaluation of search and recommender systems. He obtained his MSc. cum laude in IT-Systems Engineering at the Hasso-Plattner Institute in Potsdam, Germany. Before his PhD studies, he worked as a data scientist at Blinkist, developing multilingual recommender systems for books and podcasts for over 15M users. He was also a research assistant to Dr. Pantelis Analytis at the University of Southern Denmark (SDU). He has given a guest

lecture at the SDU, talks at Booking.com, served as a teaching assistant for an undergrad information retrieval course at the UvA, and recently published at ECIR [23] and RecSys Consequences [22].

6.3 Jin Huang

Jin Huang is a fourth-year PhD student at the University of Amsterdam (UvA) supervised by Prof. Dr. Maarten de Rijke, Dr. Herke van Hoof, and Dr. Harrie Oosterhuis. Before joining UvA, she got her BE and ME degree from the Renmin University of China. Her current research interests include but are not limited to deep reinforcement learning for recommendations, debiasing recommendations, and reinforcement learning from human feedback. She has published several research papers in top-tier conferences, including SIGIR, WSDM, and RecSys, on the topics of sequential recommendations [29, 30], debiasing recommendations [27], and reinforcement learning-based recommendations [26, 28]. Additionally, she serves as a teaching assistant for courses on information retrieval and reinforcement learning for MSc. students at the UvA.

6.4 Ali Vardasbi

Ali Vardasbi is a fourth-year PhD student in University of Amsterdam (UvA) supervised by Prof. Dr. Maarten de Rijke, and Dr. Mostafa Dehghani. He has been working on fair and unbiased learning to rank from user interactions, with an emphasis on theory. He has published several research papers in SIGIR [64, 68], CIKM [7, 65, 67], and TOIS [66], and co-organized a workshop on cross-market recommendation at RecSys 2021 [3]. He has given two invited talks about unbiased learning to rank at Elsevier and attended as a panelist in one of Elsevier's internal workshops on the fairness of ranking. He has acted as a guest lecturer for the Advanced Information Retrieval course at the UvA.

6.5 Harrie Oosterhuis

Harrie Oosterhuis is an Assistant Professor in the Institute for Computing and Information Sciences at the Radboud University Nijmegen and was until recently also a Staff Machine Learning Scientist at Twitter. His research lies on the intersection of machine learning and information retrieval and primarily concerns learning from user interactions on rankings for search and recommendation. He received his PhD cum laude from the University of Amsterdam in 2020 on the thesis titled "*Learning from User Interactions with Rankings: A Unification of the Field*" [43] under the supervision of Prof. Maarten de Rijke. He is also a recipient of the 2021 Google Research Scholar Award for early career researchers and the WSDM'21 [51], SIGIR'21 [44] and ICTIR'22 [46] best paper awards.

Harrie has extensively taught on the topic of learning to rank and unbiased learning to rank in particular, both to academic and industry audiences. Previous experience relevant to this proposal is that he has organized three tutorials on unbiased learning to rank at SIGIR 2019 [40], WWW 2020 [52], and one at the University of Queensland by special request. In addition, he has also given a four-day course on the topic at the 12th Russian Summer School in Information Retrieval [42], and lectured on this topic at MSc. level at various universities throughout the Netherlands and Europe. Finally, he has also presented on the topic to a wide international industry audience, by visiting various companies such as Google, Microsoft, Amazon, Bloomberg, and Twitter, among various others.

EXTENDED ABSTRACT

1 MOTIVATION

Learning to rank (LTR) algorithms are the cornerstone of modern search and recommender systems. Traditionally, LTR algorithms were based on supervised learning from manually-graded relevance judgments. However, obtaining relevance judgments is costly, and is often not aligned with actual user preferences [8, 59]. In contrast, click-data is cheaper to collect and is generally better aligned with user intents [32]. However, click data is usually a heavily biased signal of user preference [1, 11, 34] which the field of unbiased learning to rank (ULTR) aims to mitigate [43].

As an introduction to this field, previous tutorials [2, 40, 52] focused on introducing the fundamentals of the area to beginners and practitioners. While relevant at the time, the field of ULTR has matured significantly, and fundamental advancements have been made since then. At the time of the last tutorials, the primarily studied interaction bias was position bias [11]. Since then, the community has addressed new interaction biases, including trust bias [1], item selection bias [49], contextual bias [72, 74, 77, 78], and cascading position bias [36, 64]. For correcting biases, the method most commonly used was inverse propensity scoring (IPS). However, it is now known that IPS is not effective in correcting for all forms of interaction biases [46, 67]. Hence, several new and fundamental estimation techniques have been developed to overcome the limitations of IPS, for instance, affine-correction [51, 67] and doubly-robust estimation [47], which are a novel extension of the widely-popular IPS technique. There have also been advances in estimation techniques fundamentally different from IPS, such as two-tower models [14, 74, 78] and causal-inference-based methods [54, 63, 76]. While many ULTR methods focus on mitigating bias in historic datasets, the field of online learning to rank mitigates biases while directly interacting with users [48, 60, 75]. A recent line of work addresses both settings with methods that work offline on historical data and online, aiming to unify the field [50, 51].

Recently, LTR has also seen significant growth from the application side [1, 6, 24, 72, 78], including fair LTR [62, 73]. The focus of the previous tutorials was on the fundamentals of ULTR with a limited emphasis on practical applications. To bridge this gap between theory and practice, we plan to discuss recent applications of ULTR.

Given the advancements in the area of ULTR and its related areas, we believe it is the right time to conduct this tutorial. The field has significantly matured with recent fundamental contributions and has seen several advances in applications that are highly relevant to the IR community. While the focus of LTR was traditionally on relevance ranking, it is now commonly acknowledged that optimizing for relevance alone can result in unfairness issues [5]. In this regard, we believe that the objective of a similar area, such as fair LTR, aligns with ULTR's mission, which is to provide fair and unbiased rankings to the user. To scale up to large-scale applications, fair LTR work relies on unbiased LTR, and we hope that our tutorial will encourage further exploration in this area.

Our tutorial is expected to benefit both academic researchers and industry practitioners who are either interested in developing new ULTR solutions or applying them in real-world applications.

2 OBJECTIVES

The tutorial is based on the following two main objectives:

- To motivate and introduce the fundamental concepts of ULTR to academics or practitioners who are new to the topic.
- To provide a comprehensive overview of the important recent developments to the foundations and applications of ULTR, that is useful to both newcomers and experts in the field.

Furthermore, we aim for the following additional objectives:

- Provide the most up-to-date explanation of the mathematical foundations of the ULTR field, covering the different forms of bias that can and cannot be correct for and the latest estimation techniques. Thereby, for researchers in the ULTR field this should provide them with a strong basis to perform future research.
- Present an in-depth survey of real-world applications of ULTR, so that practitioners can have a realistic expectation of the potential impact of applying ULTR.
- Motivate and stimulate cross-disciplinary research, by enabling researchers from other areas to understand how ULTR could be useful for them. In particular, we will highlight the connection with the topic of *fairness in ranking*.

3 RELEVANCE TO THE INFORMATION RETRIEVAL COMMUNITY

The area of ULTR has grown significantly in the last couple of years, with several fundamental contributions and diverse IR applications.

The earliest tutorial in the area was by Joachims and Swaminathan [33], where they introduced counterfactual learning in the context of search and recommendation. Recent tutorials on counterfactual evaluation and learning have focused mostly on bandit feedback data [56, 57]. In the context of recommender systems, Chen et al. [9] introduce biases and debiasing strategies.

For LTR specifically, there have been tutorials introducing the offline ULTR [2, 40, 52] and online LTR [13]. However, to the best of our knowledge, no existing tutorial covers the important advancements in ULTR field that have been made in the last three years, nor their recent applications, including fair LTR.

4 FORMAT AND DETAILED SCHEDULE

The tutorial will consist of 3 hours, excluding breaks, with the following schedule:

Preliminaries (20 minutes) The first session focuses on the preliminaries; we discuss the basics of supervised LTR and some of the earliest works in position-bias and counterfactual LTR.

- **Learning to rank basics (5 minutes)**: Discuss basics of supervised LTR by introducing pointwise, pairwise, and listwise LTR methods and the concept of learning from user interactions.
- **Position bias (5 minutes)**: Discuss position bias that arises when applying traditional LTR methods on user clicks [11].
- **Counterfactual LTR (10 minutes)**: Introduce the basics of counterfactual LTR for learning from user feedback data with position bias [34].

Biases (45 minutes) In this session, we cover the recent advances in the types of interaction bias that can be tackled with ULTR methods, that go beyond the standard position bias.

- **Trust bias (10 minutes):** We discuss trust bias, where users are likely to click on items in top positions regardless of item relevance because they trust the search engine [1, 67].
- **Item Selection Bias (10 minutes):** We discuss item selection bias, where users can only interact with a fixed set of k items, and items outside the top-k position have zero chances of exposure [49].
- **Contextual Bias (10 minutes):** We discuss contextual bias, where the item's click probability is affected by its surrounding items on the display page [72, 74, 77, 78].
- **Cascading Position Bias (15 minutes):** Under the cascade model [11], the position bias of an item depends not only on the rank an item is displayed at (as many works in ULTR assume [11, 34, 70]), but also on the relevance of the items the user has inspected before [36, 64]. Thus, cascading position bias is often a more realistic assumption of user behavior, e.g., when users tend to stop searching after finding the first relevant result [11].

Novel Estimation Methods (60 minutes) The most prevalent estimation technique for correcting bias from user interaction data is IPS, introduced in the seminal works by Wang et al. [69] and Joachims et al. [34]. However, recently there have been fundamental contributions in the ULTR field with respect to novel estimation techniques. In this session, we will discuss the recent estimation techniques in detail, as per the following schedule:

- **Affine Correction Method (10 minutes):** Discuss the affine correction method introduced by Vardasbi et al. [67] as an extension to IPS, which they prove to be ineffective when correcting for trust-bias.
- **Doubly Robust Estimation (10 minutes):** Despite its popularity in the offline bandit learning literature [12, 25, 31, 55, 71], doubly robust estimation methods for position bias correction in LTR were only proposed recently [47]. We discuss this fundamental contribution to the area which overcomes some of the theoretical and practical disadvantages of IPS [47].
- **Online & Counterfactual methods (10 minutes):** Online learning methods are an alternative class of methods to counter biases in LTR [48, 60, 75], where the user preferences are learnt in an online/interactive fashion, as opposed to purely from offline data. Recently, Oosterhuis and de Rijke [51] argued that the field of ULTR can benefit from using online learning to rank via a novel online intervention-aware counterfactual estimator. Online learning has also been used to collect additional data from the logging policy to minimize the variance of the counterfactual estimate of a new ranking policy [50].
- **Marginalized propensities (10 minutes)**: IPS-based estimators rely heavily on the full-support assumption for being unbiased, i.e., the logging policy is expected to have non-zero probability over all actions assumed under the new policy. This assumption can be restrictive in a setting where the action space is large, as is typical in many real-world applications. To deal with large-action space, Saito and Joachims [58] introduced the marginal importance sampling method, which leverages so-called action embeddings to generalize to a large action space.
- **Two-tower Models (10 minutes):** So far, the focus of ULTR was primarily on identifying and developing novel bias correction methods, with limited focus on model design. Recently, with the introduction of two-tower networks, this trend has been slowly changing [14, 74, 78]. We discuss the two-tower family of correction methods.
- **Non-propensity based methods (10 minutes):** The primary estimation techniques for bias correction in ULTR are based on IPS [1, 34, 45, 67, 69]. Despite its advantages, IPS often suffers from high-variance [47]. Recently, the field of ULTR has progressed from relying primarily on IPS to other non-IPS-based bias correction techniques, such as causal-inference-based methods like Heckerman's method [53] and the Causal Likelihood Decomposition method [76].

Survey Applications (20 minutes) A significant number of papers has applied ULTR methods in different real-world settings. We discuss how the different settings that have been explored by existing work, the practical changes they require to work, and the impact they report on their performance metrics. A brief overview of the settings that will be covered:

- **Top-K ranking:** Application of ULTR in top-k rankings [1, 78].
- **Feeds recommendation:** ULTR for product feeds recommendation, with user interactions logged from a 2D grid-based user interface [72, 78].
- **Job recommendation:** Job recommendations using ULTR [10].
- **Fair LTR:** Fair policy learning for article recommendation [62].

From Unbiased to Fair LTR (20 minutes) Traditionally, fair LTR has relied on manual relevance judgments for learning fair ranking policies. Similar to the arguments in favor of click-based learning for relevance rankings, fair LTR needs to adopt click data for its widespread application. In this part of the tutorial, we discuss some applications of ULTR for fair policy learning.

- Introduce fair LTR with the traditional relevance-based judgements (10 minutes) [5, 61].
- Unbiased and fair ranking policy learning (10 minutes) [41, 73].

Conclusion and Future Work (15 minutes)

- We close by summarizing the main points discussed in the tutorial, i.e. new interaction biases from the ULTR literature, novel counterfactual estimation methods, and finally the applications of ULTR in different real-world settings.
- Recently, the emerging ChatGPT has shed new light on the use of generative models to directly generate responses to user queries. It will be crucial, but challenging, to ensure that the generated responses are unbiased, fair, and not influenced by pre-existing biases or assumptions.

5 TYPE OF SUPPORT MATERIALS TO BE SUPPLIED TO ATTENDEES

- **Slides** All presentation slides will be made publicly available on a dedicated tutorial website.
- **Bibliography** An annotated compilation of references will list all works discussed in the tutorial and should provide a good basis for further study.
- **Code** We will include references to open-source code from related work.
- **Video presentation** We plan on recording and releasing a video presentation of the tutorial.

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